3-Day Jaguar Segmentation Sprint

Project Title: Collaborative Jaguar Segmentation with U-Net

Team: All DSR Students from Batch 44

Duration: 3 Days

Mentor: Antonio Rueda-Toicen

Project Overview

This intensive 3-day sprint focuses on building a functional jaguar segmentation pipeline through collaborative teamwork. Students will manually annotate jaguar images, train a U-Net model, and benchmark its performance. Success depends equally on individual task completion and dynamic team collaboration. All work will be tracked transparently on a GitHub Kanban board.

Learning Objectives

1. Experience end-to-end ML pipeline development from annotation to evaluation
2. Practice collaborative ML workflows using version control and project management tools
3. Understand segmentation fundamentals through hands-on implementation
4. Learn effective data annotation strategies for computer vision tasks
5. Develop skills in model benchmarking and performance analysis

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Assessment Criteria (Equal Weight)

- **Individual Contributions:** Number and quality of completed tasks on the Kanban board
- **Team Collaboration:** Active communication, knowledge sharing, pair programming, and helping teammates
- **Documentation:** Clear task descriptions, progress updates, and technical decisions on GitHub

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Day 1: Setup, Annotation, and Baseline

Morning (3 hours)

Team Setup (Everyone, 30 min)

- Create GitHub repository with Kanban board
- Set up project structure and assign initial roles
- Brief: Review jaguar dataset and project goals

Infrastructure Team (2-3 students)

- Set up FiftyOne dataset for image management
- Configure CVAT for collaborative annotation
- Create data versioning structure (DVC or simple folders)
- Document setup process in README

Annotation Team (3-4 students)

- Begin manual annotation of 50-100 jaguar images in CVAT
- Focus on body segmentation masks
- Establish annotation quality guidelines
- Create annotation examples for team reference

Modeling Team (2-3 students)

- Set up PyTorch environment and dependencies
- Implement basic U-Net architecture
- Create data loaders for segmentation task
- Set up Weights & Biases for experiment tracking

Afternoon (3 hours)

Checkpoint Meeting (Everyone, 15 min)

- Share progress and blockers
- Redistribute tasks if needed

Continued Work

- Annotation Team: Continue annotation, aim for 100+ masks
- Modeling Team: Finalize U-Net implementation and test data pipeline
- Infrastructure Team: Help other teams, prepare train/val/test splits

End of Day 1 Deliverable:

- GitHub Kanban board with completed and in-progress tasks
- 100+ annotated jaguar masks in CVAT
- Functional U-Net implementation ready for training
- Data pipeline tested and documented

Day 2: Training, Iteration, and Analysis

Morning (3 hours)

Stand-up (Everyone, 15 min)

- Update Kanban board
- Identify dependencies and collaboration opportunities

Training Team (3-4 students)

- Train U-Net on annotated data
- Log experiments to W&B
- Monitor training curves and adjust hyperparameters
- Run initial inference on validation set

Annotation Team (2-3 students)

- Complete additional annotations (target: 150-200 total)
- Review and refine existing masks based on model feedback
- Create challenging test cases (occlusions, poor lighting)

Evaluation Team (2-3 students)

- Implement evaluation metrics (IoU, Dice Score)
- Create visualization scripts for predictions
- Prepare benchmarking framework

Afternoon (3 hours)

Mid-sprint Review (Everyone, 20 min)

- Demo current model performance
- Discuss quality issues and improvements

Iteration Sprint

- Training Team: Retrain with full dataset, experiment with augmentation

- Evaluation Team: Run comprehensive benchmark on test set
- All Teams: Pair programming sessions to help stuck teammates

End of Day 2 Deliverable:

- Trained U-Net model with logged experiments
- 150-200+ high-quality segmentation masks
- Preliminary benchmark results
- Updated Kanban board showing collaboration patterns

Day 3: Benchmarking, Documentation, and Presentation

Morning (3 hours)

Stand-up (Everyone, 10 min)

Final Benchmark Team (3-4 students)

- Run comprehensive evaluation on test set
- Analyze failure cases using FiftyOne
- Create performance visualizations
- Compare results across different image types (if applicable)

Documentation Team (3-4 students)

- Write comprehensive README
- Document annotation guidelines
- Create model training guide
- Prepare code for reproducibility

Presentation Team (2-3 students)

- Create presentation slides
- Prepare demo of the pipeline
- Compile key findings and visualizations

Afternoon (2.5 hours)

Final Integration (Everyone, 1 hour)

- Ensure all code is pushed and documented
- Complete Kanban board updates
- Prepare individual contribution summaries

Team Presentations (1.5 hours)

- Each team presents their work (10-15 min)
- Live demo of annotation → training → inference pipeline
- Q&A and feedback session

End of Day 3 Deliverable:

- Complete GitHub repository with:
  - Documented codebase
  - Trained U-Net model
  - Annotated dataset (150-200+ masks)
  - Benchmark results and analysis
  - Comprehensive README
- Filled Kanban board showing all contributions
- Final presentation

Kanban Board Structure

Columns

- Backlog
- To Do
- In Progress
- Review
- Done

Required Task Labels

- `annotation`
- `modeling`
- `infrastructure`
- `evaluation`
- `documentation`
- `collaboration` (for pair programming, helping others)

Task Tracking Requirements

- Assign yourself to tasks when starting
- Comment progress and blockers
- Tag teammates when you need help

- Link related issues and pull requests

Collaboration Expectations

Encouraged Behaviors:

- Asking for help publicly in team channels
- Pair programming sessions (especially cross-team)
- Code reviews before merging
- Sharing learnings and debugging tips
- Rotating through different task types
- Helping unblock teammates

Daily Rituals:

- Morning stand-up (15 min)
- Mid-day check-in (10 min)
- End-of-day sync (15 min)

Technical Stack

- **Annotation:** CVAT, FiftyOne
- **Modeling:** PyTorch, U-Net
- **Experiment Tracking:** Weights & Biases
- **Data Management:** DVC (optional) or organized folders
- **Collaboration:** GitHub (code + Kanban), Slack/Discord
- **Documentation:** Markdown, Jupyter notebooks

Success Metrics

Team Success:

- 150+ high-quality jaguar segmentation masks
- Trained U-Net model with documented performance
- Comprehensive benchmark with IoU/Dice scores
- Reproducible pipeline with clear documentation
- Active Kanban board showing distributed work

Individual Success:

- Completed 5+ meaningful tasks across the sprint
- Evidence of collaboration (code reviews, pair programming, helping others)
- Clear documentation of your contributions

Remember: This is as much about learning to work as a team on ML projects as it is about the technical implementation. Help each other, communicate openly, and document everything!